

Definition

Stability, Robustness, Resilience and Adaptability of emergent oscillations

Stability (local) of emergent oscillations means that the collective rhythmic regime of the coupled network persists under small perturbations of states or parameters and returns to its oscillatory pattern after transient disturbances.

- The rhythm survives noise.
- The rhythm survives perturbations.
- The rhythm reappears after transient disruption.
- The oscillation belongs to the attractor of the system.

Robustness means that the emergent bursting regime persists across a range of cellular and synaptic parameters.

- Bursting is not confined to a single fine-tuned parameter set.
- Variations in parameters do not eliminate the bursting regime.
- The oscillation occupies a region of parameter space.

This is parameter-space persistence of the emergent regime.

Resilience means that the network can temporarily lose or disrupt bursting under strong perturbations but subsequently reestablish the emergent slow–fast regime once the perturbation is removed.

Conceptually:

- A neuron may be silenced.
- The alternation may collapse briefly.
- But the system reorganizes and recovers bursting.
- The attractor basin is **sufficiently** broad.

This is recovery capacity after transient disruption.

Adaptability of emergent network-level bursting is the capacity of the network to systematically modify its rhythmic properties (e.g., frequency, amplitude, duty cycle) in response to modulatory input while preserving coordinated oscillatory organization.

- The rhythm does not just survive perturbations.
- It can be intentionally reshaped.
- The network changes output smoothly without losing structure.

In the swim CPG context

Adaptability means:

- Burst frequency can increase or decrease.
- Left–right coordination remains intact.
- The slow–fast regime persists.
- Modulatory input (e.g., presynaptic release rate) tunes the rhythm.

So the attractor is not destroyed — it moves continuously in parameter space.

Emergent network bursting is a system-level slow–fast oscillatory regime that belongs to the coupled swim CPG rather than to its constituent neurons.

Emergent network bursting is a composite, slow-fast dynamical regime – oscillations arising collectively through cellular–synaptic interactions in the swim CPG [as a whole] but not in its interneurons.

Network hysteresis provides the mechanism that enables this emergence bursting: slow synaptic accumulation introduces distributed memory into the network, creating *asymmetric, history-dependent states* between active, presynaptic and inactive, postsynaptic neurons. These slowly evolving synaptic perturbations reshape the fast-spiking dynamics of the neurons and drive transitions between collective activity states, thereby sustaining the network-level slow-fast bursting regime.

Network hysteresis is the mechanism that enables emergent network bursting by creating asymmetric, history-dependent states between presynaptic (driving) and postsynaptic (driven) neurons through unidirectional [chemical] synaptic perturbations.

- Network hysteresis enables emergent bursting.
- It arises from asymmetric states between driving and driven neurons.
- Synaptic currents perturb and reshape those states.
- This creates history-dependent switching.

Network oscillator: A CPG whose rhythm is an emergent property of coupling rather than an intrinsic property of its neurons.

Network oscillator stably generates network-level bursting even though its individual neurons do not burst intrinsically; when uncoupled, they exhibit only fast tonic spiking or quiescent state.

Structural stability = Behavioral stability ?

Structural stability is a property of a dynamical system: a system is structurally stable if small perturbations of its equations (parameters, functions, coupling terms) do not change its qualitative phase-space structure (e.g., number and type of attractors).

Formally:

Small C^1 perturbations preserve the topological equivalence of the system.

Behavioral stability of the swim CPG reflects the structural stability of its network-level limit cycle.

- **Structural stability** → invariance of phase-space structure under system perturbations.
- **Behavioral stability** → invariance of functional output under perturbations

The **behavioral stability** of the swim CPG arises from the structural stability of the network-level slow-fast limit cycle, which persists under small perturbations of cellular and synaptic parameters.